# MEV-Aware DeFi Liquidation: A Short Review

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***Abstract*—With strong applications in lending-protocol risk management, automated market operations, and blockchain infrastructure design, liquidation in decentralized finance (DeFi) is a fast-growing area of contemporary blockchain research. A number of liquidation techniques sought to recover undercollateralized debt using a fixed-incentive, priority-fee-driven race among liquidators. However, Maximal Extractable Value (MEV) research began to dominate the discussion once studies showed that this race raises gas expenditure and can itself extract value from borrowers. Auction and ordering mechanisms are used to structure competition and then allocate the liquidation opportunity, so mechanism design in liquidation is more crucial and challenging than it first appears. The primary goal of this paper is to review the prominence of current techniques, including both conventional priority-fee-based approaches and MEV-aware, auction-based approaches.**

***Index Terms*—decentralized finance, DeFi, liquidation, maximal extractable value, MEV, auction mechanisms.**

## I. Introduction

Decentralized lending protocols allow users to deposit collateral and borrow against it without a centralized intermediary. Liquidation is the process that keeps such protocols solvent: when a borrower's position becomes undercollateralized, a third party, called a liquidator, repays part or all of the outstanding debt and receives the collateral at a discount in return. The goal of liquidation research is to determine how this opportunity should be allocated among competing liquidators so that debt is recovered quickly without imposing unnecessary cost on the borrower or the protocol.

The problem is based on two fundamental ideas. The first is the existence of the opportunity, that is, detecting that a position has become undercollateralized. The second is allocation, that is, deciding which liquidator gets to execute it and at what cost. By treating liquidation as a transaction-ordering problem, researchers have connected it to a wider phenomenon known as Maximal Extractable Value (MEV), the profit that can be captured by controlling the order, inclusion, or exclusion of transactions within a block [1]. There is continuous empirical research, together with mechanism-design proposals, aimed at reducing the cost this competition imposes while keeping liquidation fast enough to protect protocol solvency.

As illustrated in Fig. 1, current liquidation research can be broadly classified into two categories: 1) conventional priority-fee-based approaches and 2) MEV-aware, auction-based approaches. Conventional approaches, such as fixed-spread liquidation and raw priority-gas competition, rely on the existing transaction-ordering market to decide the winner.

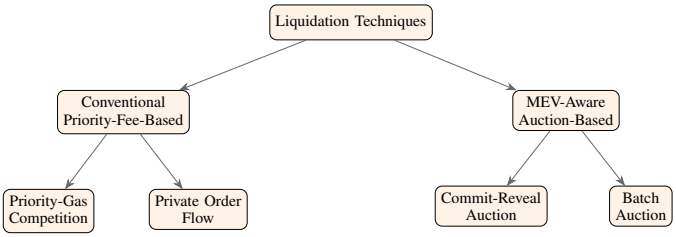



Fig. 1. Taxonomy of MEV-aware liquidation techniques.

MEV-aware approaches, by contrast, restructure the competition itself, using auctions, batching, or block-builder-aware scheduling to select a winner on efficiency grounds rather than fee speed alone. Inspired by the growing research on these two categories, this paper presents a short review of the methods commonly used for MEV-aware liquidation.

## II. Literature Review

This section places twelve representative studies side by side and compares them directly, rather than treating each as an isolated citation. The studies were chosen to span the full liquidation pipeline: empirical borrower impact in liquidations, detection of extractable value, measurement of its cost on the public mempool, structural concentration inside private order flow, and proposed fixes at the consensus and mechanism-design layers. Table I summarizes each study's focus, method, key findings, and main limitations across these four pipeline tiers. The discussion that follows groups the studies by the layer of the pipeline they address and compares their empirical and theoretical findings against one another.

**Framework versus measurement.** Daian et al. [1] and Kelkar et al. [4] sit at the conceptual end of the spectrum: the former supplies the vocabulary and systems framework that every later measurement study relies on, while the latter proposes a formal, consensus-layer fix. Neither paper measures liquidation activity directly, which is exactly the gap that Qin et al. [2] and Torres et al. [13] fill with large-scale empirical studies. Comparing the two measurement studies directly, Qin et al. focus on the economic outcome for the borrower within lending protocols specifically, whereas Torres et al. measure front-running incidents across DeFi more broadly and only report liquidation races as one contributing category. The two are complementary rather than conflicting: Qin et al. establish that the loss is economically significant per event, and Torres

TABLE I
COMPARISON OF TWELVE REPRESENTATIVE STUDIES ON MEV AND DEFI LIQUIDATION ACROSS THE TRANSACTION LIFECYCLE.

| Paper / Study | Method / Focus | Key findings & empirical metrics | Main limitations |
|---|---|---|---|
| ***A. DeFi Liquidations and Borrower Impact*** | | | |
| **DeFi Liquidations** Qin et al. [2] (2021) | Large-scale on-chain measurement | Quantifies borrower loss from priority-gas competition; static discounts (5%–10%) transfer excessive borrower equity | Measures only the existing priority-fee regime, not alternative dynamic auctions |
| **Auction Impacts** Bank of Canada [7] (2025) | Empirical and financial stability analysis | Competitive Dutch auctions cut liquidation discounts from 5.0% to 1.8% (64% reduction) and price impact by 81% | Focuses on MakerDAO collateral types; sensitive to auction duration parameter tuning |
| ***B. MEV Foundations and Priority Gas Auctions*** | | | |
| **Flash Boys 2.0** Daian et al. [1] (2019) | Systems framework, case studies | Formalized MEV and PGAs; proved uncoordinated mempool bidding induces consensus instability and time-bandit forks | Conceptual framework; offers no liquidation-specific empirical measurement |
| **Frontrunner Jones** Torres et al. [13] (2021) | Empirical detection over $\sim$11M blocks | Identified $\sim$200,000 front-running incidents ($\sim$1.8% of blocks), liquidation races among them | Detection heuristics may undercount activity routed through private builder pools |
| **Quantifying BEV** Qin et al. [3] (2022) | Empirical measurement over 32 months | Quantified $>$540M USD in extracted MEV; liquidations comprise 26% of total extracted value ($<$4.1M USD single events) | Retrospective measurement across PoW/early Merge; does not evaluate auction mechanisms |
| **SoK: Front-Running** Eskandari et al. [11] (2019) | Taxonomy of front-running attacks | Canonical taxonomy (displacement, insertion, suppression); frames priority-gas races as pure displacement attacks | Predates modern MEV terminology, PBS relays, and private mempool infrastructure |
| ***C. Proposer-Builder Separation and Private Order Flow*** | | | |
| **Flash(bot) in the Pan** Weintraub et al. [14] (2022) | Empirical measurement of private pool | Flashbots-connected miners held $>$99.9% of PoW hashing power; reveals private pools relocate rather than remove concentration | Single dominant provider studied; findings reflect the pre-Merge mining market |
| **Integrated Builders** Pai and Resnick [8] (2023) | Auction-theoretic modeling | Integrated searcher-builders gain zero-latency bundle visibility, subjecting independent searchers to winner's curse | Stylized theoretical model; does not evaluate empirical on-chain liquidation traces |
| **Private Order Flows** Wang et al. [9] (2024) | Empirical and game-theoretic analysis | Documents bidding monopolies: top 3 block builders assemble $>$95% of blocks via exclusive private flow | Concentration trajectory depends on evolving PBS market rules and relay dynamics |
| **Artificial Latency** Natale and Moser [10] (2023) | Empirical measurement of PBS timing | Proposers systematically delay block proposals by 500–1000 ms to extract a median 1.28% increase in MEV bribes | Measures consensus proposer timing games rather than application-layer auction clearing delays |
| ***D. Order-Fairness and Mechanism Countermeasures*** | | | |
| **Order-Fairness (Aequitas)** Kelkar et al. [4] (2020) | Protocol design, formal proofs | Defines order-fairness guarantees (Aequitas), proving standard consensus does not prevent transaction reordering | High communication complexity; lacks liquidation-specific evaluation or latency cost estimate |
| **SoK: Reordering** Heimbach and Wattenhofer [15] (2022) | Systematization of knowledge (SoK) | Formulates the MEV trilemma: no surveyed countermeasure satisfies low latency, manipulation resistance, and decentralization | Qualitative comparative survey; introduces no primary empirical liquidation datasets |

et al. establish that such events are not rare, at roughly 1.8% of all blocks in their sample (Fig. 2).

**Public mempool versus private order flow.** Torres et al. [13] and Weintraub et al. [14] measure the same underlying phenomenon, MEV extraction, at two different points in the pipeline: the former in the public mempool, the latter inside a single private-pool operator. Read together, their results suggest that moving competition off the public mempool does not remove concentration, it relocates it: Weintraub et al. found over 99.9% of hashing power connected to one private-pool provider during the period studied (Fig. 3), and Wang et al. [9] later found a similar concentration pattern one layer down, among block builders rather than miners. This agreement across two independent studies, conducted on different infrastructure and in different years, is stronger evidence for structural concentration than either result would be alone.

**Design proposals versus their own trade-offs.** Kelkar et al. [4] and Heimbach and Wattenhofer [15] both address the allocation side of the problem rather than measuring it, but at different layers: Kelkar et al. redesign the consensus protocol itself, while Heimbach and Wattenhofer systematize application-layer and mechanism-level countermeasures, commit-reveal and batch auctions among them. The SoK's central finding, that no surveyed scheme achieves low latency, manipulation resistance, and decentralization simultaneously,

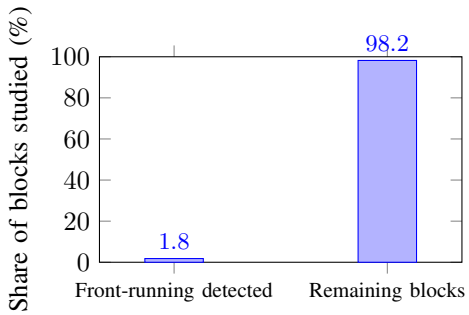



Fig. 2. Share of the ∼11 million Ethereum blocks studied by Torres et al. [13] that contained at least one detected front-running incident (∼200,000 incidents in total).

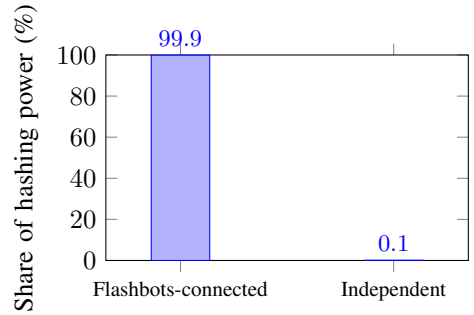



Fig. 3. Share of Ethereum hashing power connected to Flashbots during the period measured by Weintraub et al. [14].

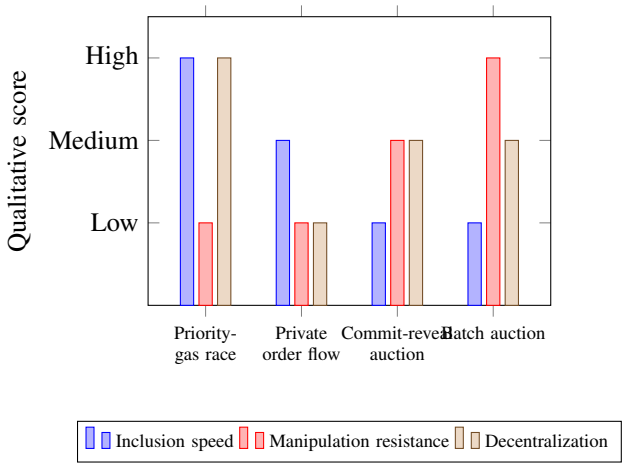



Fig. 4. Qualitative comparison of the four mechanisms in Table II, illustrating the trade-off Heimbach and Wattenhofer [15] identify across surveyed countermeasures. Scores are illustrative summaries of the discussion in this section, not measured data.

applies directly to the four mechanisms compared in Table II and is visualized qualitatively in Fig. 4: priority-gas racing wins on latency but not manipulation resistance, while commit-reveal and batch auctions trade latency away for the other two properties.

## III. Conventional Priority-Fee-Based Approaches

According to Qin et al. [2], liquidation outcomes can be viewed at three levels: incentive design, execution cost, and borrower impact. Incentive design fixes the discount a liquidator receives; execution cost depends on the gas market at the moment of liquidation; and borrower impact depends on how much collateral is lost relative to the debt recovered. These three levels are used to model most conventional liquidation mechanisms in use today. The foundational work of Daian et al. [1] applied a unified framework to describe front-running and reordering risk across decentralized exchanges and lending protocols alike. Competition for liquidation was found to operate in two ways: 1) mempool-visible priority racing and 2) private-channel submission. In the following subsections, these techniques are covered in detail.

### A. Priority-Gas Competition

A group of researchers [2] has studied how liquidators bid gas prices against each other once a position becomes eligible, a process that mirrors a first-price, continuous-time auction over block space. The liquidator who is willing to pay the highest priority fee usually wins inclusion, so the discount margin is competed away toward the cost of the marginal gas bid. The empirical study [2] found that this competition is real and economically significant, but it also found that borrowers can lose substantially more value than the protocol strictly needs to recover, particularly during periods of high volatility. In a large-scale measurement study, Torres, Camino, and State [13] found close to 200,000 front-running incidents across roughly 11 million Ethereum blocks, showing that priority-gas competition is not a theoretical concern but a persistent, high-volume feature of the network, of which liquidation races form a significant share.

Front-running and transaction-reordering studies [1] summarized this competition as an instance of a broader transaction-ordering market. Existing block-construction incentives were used to explain why this market persists even as tooling around it matures. Eskandari, Moosavi, and Clark [11] systematized front-running attack types on blockchains before the MEV terminology was widely used; their taxonomy predates most of the liquidation-specific literature but still underlies how later studies classify priority-gas competition. According to this line of work, a single gas bid cannot adequately capture a liquidator's true willingness to pay, particularly when several liquidators are competing for the same opportunity. Thus, later designs needed a richer signal than a single priority fee to allocate the opportunity efficiently.

### B. Private Order Flow and Direct Submission

An alternative to the public mempool race is to submit liquidation transactions through private order flow, bypassing the public transaction pool to avoid front-running by other searchers. Such approaches shift competition away from the mempool and toward whichever block builder or relay accepts the private transaction. Private order-flow arrangements can reduce visible competition, but they also concentrate market

power in the hands of the builders who receive this flow [9]. Weintraub et al. [14] measured this concentration directly for Flashbots, the most widely used private-pool infrastructure, and found that Flashbots-connected miners accounted for over 99.9% of Ethereum's hashing power during the period studied, even though Flashbots was designed to reduce, not concentrate, the negative externalities of MEV competition. When too little of the liquidation flow is public, the effective market for block space becomes harder to observe and to reason about, while relying entirely on public bidding can leave liquidators exposed to being outbid at the last moment. Pai and Resnick [8] extend the study of this trade-off by examining how integrated builders gain structural advantages over independent ones under this regime.

Priority-fee-based approaches produce results quickly and require no protocol changes, but they are not necessarily efficient. Bidding becomes wasteful as gas prices climb during volatile markets, since most of the extra fee paid does not go toward recovering debt. It can be costly and unpredictable for liquidators to participate under these conditions, which makes the mechanism less attractive for smaller searchers.

## IV. MEV-Aware and Auction-Based Approaches

Because of their ability to separate the question of who should win from the question of how fast a transaction is included, auction mechanisms are increasingly proposed for liquidation. An auction-based mechanism collects bids or commitments from liquidators and selects a winner using an explicit scoring rule rather than raw transaction-inclusion speed. Heimbach and Wattenhofer [15] systematized the wider space of reordering-manipulation countermeasures in DeFi and found that, across the schemes surveyed, none fully satisfied every property a liquidation mechanism would want at once, namely low latency, resistance to manipulation, and minimal impact on decentralization. This trade-off motivates much of the auction-specific work reviewed below. Kelkar et al. [4] developed order-fairness properties for the underlying consensus layer to reduce a single party's ability to manipulate transaction order for profit. It was followed by mechanism-design work aimed at applying similar fairness guarantees specifically to liquidation auctions. Deeper redesigns of the ordering layer are prone to introducing new latency costs of their own. Natale and Moser [10] quantified the cost of artificially delaying block proposal within a proposer-builder separation (PBS) context, a cost that any batching or commit-reveal liquidation auction must budget for. Ethereum.org's technical documentation [12] provides a continuously updated, practitioner-facing description of how PBS, relays, and builders fit together, and is a useful companion to the academic sources above for readers unfamiliar with the current infrastructure, though as a living document it should be read alongside the dated academic sources rather than in place of them.

Werner et al. [5] and Gogol et al. [6] suggested a practical way of situating liquidation-specific mechanisms within the broader systematization of DeFi risk. The studies demonstrated that liquidation cannot be optimized in isolation from other MEV-generating activities such as arbitrage, since the same searchers and the same block-builder market participate in both.

TABLE II
Advantages and disadvantages of MEV-aware liquidation mechanisms.

| No. | Mechanism | Advantages | Disadvantages |
|---|---|---|---|
| 1. | Priority-gas race | Simple, requires no protocol change | Wasteful bidding, unpredictable cost for liquidators |
| 2. | Private order flow | Reduces visible front-running | Concentrates power in a few builders |
| 3. | Commit-reveal auction | Scores bids on efficiency, not just speed | Reveal delay adds latency cost |
| 4. | Batch auction | Aggregates competing bids into one clearing event | Batching window may allow price to move against the protocol |

Wang et al. [9] proposed a builder-bidding-dynamics analysis for Ethereum's block-building market that could help predict which liquidation-auction designs remain viable as private order flow grows. The analysis, including bidding concentration among a small number of builders, was used to explain the direction of the market. It suggested caution for real-time auction designs that depend on competitive, decentralized block building. When working on auction-based liquidation, the Bank of Canada working paper [7] took a different approach compared to the purely technical studies, evaluating liquidation mechanisms and price impact from a financial-stability standpoint. The macro-level framework decreased the reliance on purely searcher-level metrics. Instead, it used protocol-level solvency outcomes that are less specific to any one implementation.

Qin, Zhou, and Gervais [3] highlighted the limitations of narrow, single-metric measurement for extractable value more broadly, in the context of contemporary MEV research. By quantifying how much value remains extractable under different market conditions, the study helped establish a benchmark against which auction-based liquidation designs can be judged. It substantially improved the ability to compare designs on a common empirical basis. Table II shows the advantages and disadvantages of auction-based mechanisms relative to the priority-fee baseline.

### A. Challenges in MEV-Aware Mechanism Design

Although the positive aspects of auction-based liquidation for reducing execution cost have so far been demonstrated in isolated studies, significant work is still needed to address the apparent problems. Real searcher behavior has a significant influence on the accuracy of any mechanism evaluated only in theory. Mechanisms are usually judged using historical, priority-fee-era data, but there is a scarcity of data on how strategic liquidators would actually bid under a redesigned auction. Model mismatch and an unreliable efficiency estimate are caused by this gap between theory and observed behavior.

Latency is the main obstacle when introducing a commit-reveal or batching step. Faster settlement results from shorter reveal windows, while a more accurate, less manipulable clearing price results from longer ones. A small share of liquidation events is frequently responsible for most of the borrower loss recorded in existing datasets. The remaining events, however, are part of the low-volatility, low-competition regime. High importance is given to these small, high-loss events for evaluation purposes, which may also be taken into account when tuning a mechanism. Nevertheless, a design tuned only toward the least common, highest-loss events risks being biased away from the ordinary case that makes up most liquidation volume.

## V. Conclusion

In blockchain-based lending, protocol design favors mechanisms that recover debt with more efficiency and less collateral loss than a raw priority-fee race allows. We frequently overlook a number of additional costs that are probably imposed on borrowers but may not be visible in liquidator profit figures alone, because they are distributed across many small events. MEV-aware liquidation is intended for both public mempool and private order-flow environments. A liquidation event is shaped by weaving together several forces: liquidator incentive, gas market conditions, block-builder behavior, and borrower risk. To evaluate mechanisms across these forces, isolate the effect of any one design change, and ultimately recommend a mechanism, metrics like execution cost, borrower loss, and inclusion latency are needed together rather than one at a time.

Following the growth of proposer-builder separation and private order flow, MEV research began to move beyond the priority-fee model that defined early DeFi liquidation. By taking advantage of structured bidding and explicit scoring rules to allocate the liquidation opportunity, auction-based mechanisms were proposed to improve the protocol's cost profile. Different auction-based liquidation proposals are rich in reducing wasted gas expenditure, tightening the link between bid and true willingness to pay, and, in principle, improving borrower outcomes, whereas the priority-fee baseline was built on raw transaction-inclusion speed alone. In liquidation-mechanism design, commit-reveal and batch auctions are increasingly studied. However, no single existing study jointly evaluates liquidator strategy, execution cost, borrower loss, protocol recovery, and block-builder interaction within one consistent experimental framework, which remains an open problem for future work.

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